

A Comparative Study on ESG Practices in Selected Indian Corporates

Done by:

SHASHANK TIWARI

St. Xavier's College (Autonomous), Kolkata

Project Guide / Mentor Name: Dr. Piyali Sengupta

Period of Internship: 14th Jan 2025 - 30th April 2025

Report submitted to: IDEAS – Institute of Data
Engineering, Analytics and Science Foundation, ISI
Kolkata

DECLARATION

I hereby declare that the report titled “A Comparative Study on ESG Practices in Selected Indian Corporates” has been prepared as part of my industrial internship under the guidance of Ms. Piyali Sengupta, at the IDEAS-TIH, Indian Statistical Institute (ISI), Kolkata.

This report is a result of my sincere efforts and independent academic work carried out during the internship period. The study is based entirely on secondary research, involving the collection and analysis of publicly available data from official sources such as company websites, annual reports, financial statements, and Business Responsibility and Sustainability Reports (BRSRs) for the financial years 2021–2022, 2022–2023, and 2023–2024.

I affirm that all the data sources used in this report have been duly acknowledged and cited, and that the findings, interpretations, and conclusions presented are purely academic and are intended for educational and research purposes. No part of this report has been copied or reproduced from any other existing work or project in any manner that would constitute a violation of academic integrity.

I express my deepest gratitude to Ms. Piyali Sengupta for her valuable guidance, consistent support, and insightful feedback throughout the course of this project. We are also thankful to IDEAS-TIH, ISI Kolkata for providing us with the opportunity and platform to undertake this research.

A sincere thank you to my faculty mentor Mr. Siladitya Mukherjee for his continued guidance throughout this project.

I would also like to thank my teammates at IDEAS with whom I collaborated in the completion of the internship.

- Sidharth Singh- B.A. Business Economics (H), College of Vocational Studies, University of Delhi
- Sowmika Rao Samudrala- B.Tech, IIT Mandi, Himachal Pradesh.
- Sharanya Chakraborty- BSc. Economics (H), Seth Anandram Jaipuria College, Kolkata

Submitted with utmost honesty and academic responsibility.

Contents

1. INTRODUCTION
 - 1.1. Environmental Aspect
 - 1.2. Social Aspect
 - 1.3. Governance Aspect
 - 1.4. Interconnection of ESG Components
 - 1.5. Why Does ESG Matter?
 - 1.6. How do the Investors Use ESG to Guide Sustainability?
2. OBJECTIVES AND KEY FOCUS AREA
 - 2.1. Analysis of ESG Performance Across Companies
 - 2.2. Economic Impacts of ESG Adoption in Corporate Business
 - 2.3. Role of Analytical Tools and Frameworks in ESG Management
 - 2.4. Interlinking Data Analysis and the Economic Rationale for ESG
3. BRSR REPORTING, GRI, AND ESG PARAMETERS: AN OVERVIEW
 - 3.1. The Rise of ESG and the Need for Reporting
 - 3.2. Business Responsibility and Sustainability Reporting (BRSR)
 - 3.3. Global Reporting Initiative (GRI)
 - 3.4. ESG Parameters: The Building Blocks of Reporting
 - 3.4.1. Energy Consumption
 - 3.4.2. Energy Intensity
 - 3.4.3. Greenhouse Gas (GHG) Generation
 - 3.4.4. Spend on Research and Development (R&D)
 - 3.5. Interconnectedness and the Path Forward
4. METHODOLOGY (CASE STUDY & STATISTICAL TOOLS)
 - 4.1. Case Study Method: A Comparative Structure
 - 4.2. Statistical Tools and Quantitative Analysis
 - 4.3. Rationale for Mixed-Method Design
 - 4.4. Limitations of Methodology
5. DATA COLLECTION
 - 5.1. Sample Summary
6. ESG SCORES
7. DATA ANALYSIS AND RESULTS
 - 7.1. How data analytics benefit ESG reporting and monitoring
 - 7.2. Challenges in Integrating Data Analytics into ESG Reporting
 - 7.3. Consistency of the data procured
 - 7.4. Analysis of the data procured
8. RECOMMENDATIONS, OUTCOME
9. CONCLUSION
10. REFERENCES

Introduction

ESG, short for Environmental, Social, and Governance, is a framework used to assess the sustainability and ethical impact of a company. An ESG framework is essential for assessing companies by investors, consumers, and stakeholders. It goes beyond traditional financial metrics to evaluate how a company performs in areas that affect society and the environment. This approach helps investors make informed decisions that align with their values and long-term goals.

Environmental aspect:

The environmental aspect focuses on how a company impacts the natural world. It assesses practices that mitigate harm to ecosystems, conserve resources, and combat climate change. Key areas include:

- **Climate Change and Carbon Emissions:** Companies are evaluated on their greenhouse gas (GHG) emissions, energy efficiency, and commitments to net-zero targets. Firms with sustainable energy initiatives and carbon-reduction strategies often rank higher in ESG ratings.
- **Resource Management:** Investors examine how organizations manage water, waste, and raw materials. For example, businesses adopting circular economy models—where resources are reused and recycled—are viewed favorably.
- **Biodiversity and Land Use:** Companies involved in deforestation or habitat destruction may score poorly, while those supporting biodiversity conservation and sustainable agriculture receive better ESG evaluations.
- **Renewable Energy and Innovation:** Transitioning to renewable energy sources like wind or solar power and investing in green technologies are considered essential for future resilience.

Companies demonstrating strong environmental stewardship are not only reducing risks but also seizing opportunities in the growing green economy.

Social Aspect:

The social pillar evaluates how a company interacts with employees, customers, suppliers, and communities. This aspect emphasizes the importance of ethical and equitable practices that contribute to societal well-being.

- **Diversity and Inclusion:** Companies with diverse leadership teams and inclusive workplace cultures often outperform those lacking representation, as diversity fosters innovation and broadens perspectives.
- **Labour Practices and Human Rights:** Ethical labour practices, including fair wages, safe working conditions, and respect for human rights across supply chains, are critical for ESG ratings.

- **Community Engagement:** Businesses that actively support local communities through philanthropy, volunteering, or infrastructure development demonstrate social responsibility.
- **Customer Relations:** Maintaining transparent, honest interactions with consumers and ensuring data privacy and product safety are key indicators of strong social governance.

Socially conscious companies build trust, foster loyalty, and create positive impacts, all of which contribute to long-term success.

Governance aspect:

Governance addresses the internal structure and leadership of an organization, focusing on accountability, transparency, and ethical decision-making.

- **Board Diversity and Independence:** A well-structured board with diverse perspectives and independent oversight is essential for avoiding conflicts of interest and ensuring balanced decision-making.
- **Executive Compensation:** Transparent and performance-linked executive pay structures align leadership incentives with shareholder interests.
- **Anti-Corruption and Ethical Practices:** Companies with strong policies against bribery, fraud, and unethical behavior demonstrate robust governance.
- **Shareholder Rights:** Ensuring shareholders have a voice in major decisions, such as mergers or policy changes, reflects good governance.

Governance forms the backbone of a company's ESG strategy, ensuring that environmental and social goals are met with integrity.

Interconnection of ESG components:

While each component has distinct criteria, they are deeply interconnected. For instance, a company with poor governance may face social and environmental controversies, impacting its reputation and financial performance. Conversely, a business excelling in all three ESG areas is better equipped to adapt to global challenges, ensuring sustainable growth.

Why Does ESG Matter?

ESG investing has emerged as more than just a trend; it is a transformative approach that aligns financial growth with global sustainability goals. Understanding why ESG matters requires examining its influence on sustainable development, investor values, and risk management.

How do the Investors Use ESG to Guide Sustainability?

Investors use ESG (Environmental, Social, and Governance) criteria as a framework to assess a company's long-term sustainability and ethical impact, which helps guide responsible investing.

Investors evaluate how companies manage environmental risks like carbon emissions, climate change, pollution, and resource depletion. Companies with poor environmental practices may face regulatory fines, lawsuits, or reputational damage. Investors also analyse how a company treats workers, interacts with communities, and manages its supply chain. Poor labour practices or human rights violations can result in boycotts or legal action. Weak corporate governance (e.g., lack of board diversity, corruption, poor transparency) can lead to mismanagement and scandals. Sustainable companies tend to have better long-term prospects. Investors use ESG to align with companies that are forward-thinking, resilient to social and environmental changes, and better managed.

Investors who apply ESG criteria believe that these factors can provide critical insights into a company's long-term prospects, making companies with strong ESG practices, more resilient and potentially more profitable investments. ESG investing has gained momentum as more investors seek not only financial returns but also to contribute to positive societal outcomes. This trend reflects a growing belief that sustainable, responsible business practices are linked to a company's overall performance and risk profile. ESG plays a critical role in promoting sustainable development. By encouraging responsible business practices, Environmental, Social, and Governance helps address global challenges such as climate change, social inequality, and governance failures. This contribution is essential for creating a sustainable future for all.

Objectives and Key Focus Area

This study aims to develop a sustainability framework to evaluate the overall sustainability performance of organizations. It focuses particularly on a set of organizations, assessing their performance across three core dimensions: Environmental, Social, and Governance (ESG). The report presents a comprehensive review of the sustainability status of these enterprises and examines relevant literature to build a conceptual understanding of corporate sustainability and related constructs. The study is based on secondary data analysis and concludes by summarizing the findings in light of the research objectives.

The major areas of focus of this study are fourfold:

1. Analysis of ESG Performance Across Companies

The primary focus is on the analysis of ESG performance by the chosen companies. This encompasses a detailed understanding of how every company has woven ESG philosophies into the corporate culture, policies, and operational structures. The research has a detailed perspective on the environmental programs—carbon neutrality targets, green investments in renewable energy, and mechanisms for emission controls—undertaken by each company. It also explores the social aspects like employee well-being, community engagement, occupational health and safety policies, and diversity and inclusion practices. Governance, the third component, is assessed through an examination of transparency, ethical behavior, anti-corruption policies, board diversity, and shareholder communication practices.

This part is not simply a description of ESG initiatives but a critical examination of their effectiveness, implementation, and compatibility with international standards of sustainability such as the Global Reporting Initiative (GRI), the UN Sustainable Development Goals (SDGs), and India's environmental and social goals.

2. Economic Impacts of ESG Adoption in Corporate Business

Although ESG is usually viewed through the lens of a compliance or normative approach, this research highlights its economic and strategic facets. One of the key goals is to measure the tangible and intangible advantages. They include better access to capital, enhanced reputation, reduction of risk, and long-term profitability. For instance, firms with robust ESG profiles tend to have positive credit ratings and more institutional investor trust. Additionally, ESG performance is increasingly becoming a stock market resilience determinant in turbulent economic environments.

This research seeks to draw linkages between ESG initiatives and financial metrics—such as Return on Assets (ROA), Return on Equity (ROE), and market valuation—highlighting how sustainable business practices can align with, and even accelerate, economic performance. The study also explores cost-saving mechanisms embedded in environmental efficiency measures, as well as the economic value of social goodwill.

3. Role of Analytical Tools and Frameworks in ESG Management

The third major theme delves into the function of data, metrics, and analytics tools in driving ESG integration and performance monitoring. With today's digital world, ESG is no longer a qualitative story—it is now more data-intensive. This research seeks to discover how firms are leveraging measurable Key Performance Indicators (KPIs), dashboard tools, ESG rating frameworks, and AI-powered platforms to assess and report their sustainability stories.

Are the tools merely being deployed for surface reporting, or are they influencing strategic choices? How is ESG data integrated into risk modelling or supply chain management? These are but a few of the questions this study attempts to respond to. It also assesses third-party ESG rating approaches, materiality evaluation frameworks, and real-time monitoring solutions being utilized by the firms under consideration.

By emphasizing this dimension, the research highlights the confluence of sustainability, corporate acumen, and technological innovation—a convergence that is dynamically unfolding in international business.

4. Interlinking Data Analysis and the Economic Rationale for ESG

The ultimate point of attention combines the other aspects by looking at how use of data analysis aids the financial justification of ESG. Because so much structure and unstructure data is collected through sustainability reports, sophisticated methodologies like predictive model, trend measurement, and simulation of scenarios more and more tend to be utilised to predict the risks as well as reward of ESG investments.

This section discusses how data analysis helps identify ESG-related risks (such as regulatory fines or climate disruptions), estimate future sustainability patterns, and highlight high-impact investment areas. The research takes into account whether or not these firms are leveraging data for compliance alone or as an actual catalyst of innovation and competitive differentiation. It also examines the function of real-time dashboards, ESG scenario planning, and natural language processing tools in informing ESG strategy.

Through this investigation, the study highlights how ESG has grown from a rigid reporting framework to a dynamic, data-driven field with real business consequences.

Collectively, these four areas of focus facilitate a multi-faceted evaluation of ESG practices among some of India's most powerful corporate players. This research does not seek to make value judgments but to shed light on patterns, challenges, and opportunities in ESG integration—thus adding to the emerging academic and policy literature on sustainable corporate governance in India.

BRSR Reporting, GRI, and ESG Parameters: An Overview

With a world marked by growing environmental awareness, social responsibility, and the need for open governance, the term Environmental, Social, and Governance (ESG) has shifted from the periphery to the center of business operations and investment practices. Stakeholders such as investors, customers, employees, and regulators are increasingly considering companies not only on the basis of their financials but also on how they contribute to the well-being of the planet and its inhabitants. This has given rise to several reporting frameworks and parameters designed to standardize and quantify ESG performance. Among the most important frameworks and principles within this changing landscape are Business Responsibility and Sustainability Reporting (BRSR), the Global Reporting Initiative (GRI), and the wide range of ESG parameters they include.

The Rise of ESG and the Need for Reporting

The incorporation of ESG considerations in business decision-making recognizes that the creation of value over the long term is closely associated with sustainable behaviour and ethical practices. Organizations that can properly manage their ecological footprint, have good social relations, and obey strong governance practices tend to better manage risk, leverage opportunity, and create resilience while surrounded by international pressures such as climate change, inequality, and the scarcity of resources.

However, in order to really know and compare organizations' ESG performance, there needs to be standardized and clear reporting systems. This is where reporting frameworks such as BRSR and GRI step in, which offer blueprints to businesses on how to release their related ESG information in a thorough and comparable way.

Business Responsibility and Sustainability Reporting (BRSR)

The Business Responsibility and Sustainability Report (BRSR) is a reporting framework mandated by the Securities and Exchange Board of India (SEBI) for the top 1,000 listed companies in India by market capitalization, starting from the financial year 2022-23. It represents a significant step towards mainstreaming ESG reporting in the Indian context, evolving from the earlier Business Responsibility Reporting (BRR) framework.

The BRSR is designed to provide a more quantifiable and standardized disclosure of a company's performance against the nine principles of the "National Guidelines on Responsible Business Conduct" (NGBRCs). These principles cover a wide range of ESG aspects, including ethics, product sustainability, employee well-being, human rights, community engagement, and environmental protection.

The BRSR framework is structured into three key sections:

1. **General Disclosures:** This section requires basic information about the company, its operations, and its approach to sustainability.

2. **Management and Process Disclosures:** This delves into the company's policies, processes, and governance mechanisms related to the NGBRC principles.
3. **Principle-wise Performance Disclosures:** This is the core of the BRSR, requiring companies to report on specific performance indicators related to each of the nine NGBRC principles. These indicators are a mix of essential (mandatory) and leadership (voluntary) disclosures, encouraging a progressive adoption of comprehensive reporting.

The BRSR aims to enhance transparency, enable better stakeholder understanding, and promote responsible business conduct among Indian companies. By aligning with global reporting frameworks like GRI and the Task Force on Climate-related Financial Disclosures (TCFD), it also facilitates international comparability and investor confidence.

Global Reporting Initiative (GRI)

The Global Reporting Initiative (GRI) is an international nongovernmental organization that offers a globally accepted framework for sustainability reporting. GRI Standards constitute a modular structure including Universal Standards for all organizations and Topic Standards covering particular economic, environmental, and social subjects.

The GRI framework is used across all sizes and sectors of companies globally to report on their effects on the economy, environment, and society. It focuses on the notion of materiality, and companies are obligated to report on issues that are their most material impacts and are most relevant to stakeholders.

The GRI Standards are structured into three series:

1. **GRI 100 Series (Universal Standards):** These foundational standards provide guidance on how to use the GRI framework and report general information about the organization, its context, and its material topics.
2. **GRI 200 Series (Economic Topics):** These standards cover a range of economic impacts, including economic performance, market presence, indirect economic impacts, procurement practices, and anti-corruption.
3. **GRI 300 Series (Environmental Topics):** This series focuses on environmental impacts such as materials, energy, water and effluents, biodiversity, emissions, effluents and waste, environmental compliance, and transport.
4. **GRI 400 Series (Social Topics):** These standards address social impacts related to labour practices, human rights, society, and product responsibility.

The GRI Standards promote transparency and accountability by providing a structured and standardized way for organizations to disclose their sustainability performance, enabling stakeholders to make informed decisions.

ESG Parameters: The Building Blocks of Reporting

ESG parameters are the individual metrics and indicators that firms utilize to quantify and report their environmental, social, and governance performance. They can be qualitative or quantitative and give insights into a firm's risks, opportunities, and overall sustainability footprint.

Both BRSR and GRI frameworks have a broad array of ESG parameters. Whereas BRSR parameters follow the nine NGBRC principles, GRI's Topic Standards provide a more specific and elaborate list of possible disclosures in economic, environmental, and social areas.

To illustrate the nature of ESG parameters, let's define some specific examples:

1. **Energy Consumption:** This refers to the total amount of energy used by an organization in its operations over a specific period. It is typically measured in kilowatt-hours (kWh), megajoules (MJ), or other relevant energy units. Reporting on energy consumption helps stakeholders understand a company's operational efficiency and its reliance on different energy sources, including fossil fuels and renewables. High energy consumption can indicate potential inefficiencies and a larger carbon footprint.
2. **Energy Intensity:** Energy intensity measures the amount of energy consumed per unit of output or activity. It provides a normalized metric to assess energy efficiency. For example, it can be expressed as energy consumed per unit of revenue, per unit of production, or per square meter of building space. A lower energy intensity indicates that a company is using less energy to generate its output, signifying improved energy efficiency. Tracking energy intensity trends over time can reveal a company's progress in reducing its energy footprint.
3. **Greenhouse Gas (GHG) Generation:** This parameter refers to the total amount of greenhouse gases emitted directly and indirectly by a company's activities. GHGs, such as carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O), contribute to climate change. Reporting on GHG emissions typically includes Scope 1 emissions (direct emissions from owned or controlled sources), Scope 2 emissions (indirect emissions from the generation of purchased electricity, steam, heating, and cooling), and increasingly Scope 3 emissions (all other indirect emissions that occur in a company's value chain). Quantifying and reporting GHG generation is crucial for understanding a company's contribution to climate change and its efforts towards decarbonization.
4. **Spend on Research and Development (R&D):** While R&D expenditure is often viewed as a financial metric, it can also be a significant ESG parameter, particularly when considered in the context of sustainability. Investments in R&D can drive innovation in cleaner technologies, resource-efficient processes, and sustainable products and services. Reporting on the proportion of R&D spending directed towards sustainability-related projects can provide insights into a company's commitment to long-term sustainability and its potential to contribute to a greener economy. This parameter links governance (strategic allocation of resources) with environmental and potentially social outcomes.

Interconnectedness and the Path Forward

BRSR, GRI, and various ESG parameters are interconnected elements in the broader sustainability reporting ecosystem. While BRSR provides a specific regulatory mandate and framework for Indian companies, GRI offers a globally applicable and comprehensive set of standards. Both frameworks rely on the disclosure of relevant ESG parameters to enable meaningful assessment of corporate sustainability performance.

As the focus on ESG continues to intensify, the quality, comparability, and reliability of reported data will become even more critical. Efforts are underway globally to further standardize ESG reporting, enhance data quality, and ensure greater consistency across different frameworks.¹ This will empower stakeholders with the information they need to make informed decisions and drive the transition towards a more sustainable and responsible global economy.² Companies that embrace comprehensive ESG reporting, utilizing frameworks like BRSR and GRI and diligently tracking and improving their ESG parameters, will be better positioned for long-term success in this evolving landscape.

Methodology (Case Study & Statistical Tools)

The methodology used in this study combines qualitative and quantitative methods, with a strong emphasis on real-life case studies and statistical analysis to draw meaningful conclusions regarding the ESG practices of six large Indian companies: NHPC, NTPC, JSW Group, Reliance Industries, Tata Power Ltd., and Adani Power Ltd. These companies were chosen based on their visibility in India's industrial and energy space, their diversified ownership patterns (public sector and private sector), and their publicly available ESG information.

This section describes the two-pronged methodological approach adopted in the research—(1) case study analysis to measure the implementation of ESG at the firm level, and (2) statistical methods to quantify, contrast, and interpret the ESG-economic impact relationship.

A. Case Study Method: A Comparative Structure

Case study method is the qualitative basis of this study. Using this method, each of the companies was analyzed separately in order to construct a comprehensive picture of its ESG journey, strategic decisions, and sectoral issues. The comparative structure of this method enabled us to examine ESG practices not separately but comparatively with each other—identifying patterns, divergences, and sectoral dynamics.

Case studies were crafted using publicly accessible documents such as:

1. Annual reports and financial reports (FY 2021–2023)
2. Business Responsibility and Sustainability Reports (BRSR)
3. ESG disclosures filed with regulatory bodies (e.g., SEBI)
4. Official sustainability reports conforming to GRI and UN SDG frameworks.
5. Industry publications, and corroborated investor presentations

For every company, we examined ESG-related initiatives within three categories:

1. Environmental (E): Emissions reduction policies, clean energy usage, resource conservation, biodiversity, and climate change adaptation.
2. Social (S): Employee health and well-being, occupational safety, gender balance, community, and education or health outreach.
3. Governance (G): Board make-up, shareholders' rights, audit processes, ethical codes, transparency, and anti-corruption.

These sectors were not evaluated as discrete categories but were conceptualized in terms of their linkages and positioning in larger corporate strategy. For example, Reliance Industries' forays into green hydrogen were not considered just as environmental innovations but also examined in terms of possible social spillovers (creation of jobs) and governance issues (alignment with regulation).

In addition, the strategy of each firm regarding stakeholder engagement and ESG risk management was also reviewed to determine not only what policies existed, but how well they were integrated into core business practices.

B. Statistical Tools and Quantitative Analysis

To supplement the qualitative findings obtained from the case studies, the quantitative approach was applied to establish empirical relationships and derive objective measures of ESG impact. Statistical methods were utilized to accomplish this, with specific emphasis on the interrelations between ESG indicators and financial or operational performance.

The most important steps in statistical analysis were:

- **ESG Score Normalisation:** As rating agencies employ various scales, all ESG scores were normalized to a scale from 0 to 100 so that meaningful cross-company comparisons could be made.
- **Trend Analysis:** Time-series analysis of FY 2020 to FY 2023 ESG disclosures was performed to analyze the trend of ESG commitment over time. This helped us identify whether ESG practices were reactive (compliance-driven) or proactive (strategy-driven).

Rationale for Mixed-Method Design

A mixed-method approach was used for this research in order to counter the limitations inherent in any one methodological paradigm. Whereas quantitative methods assist in the identification of measurable patterns and the inference of causality, qualitative case studies provide the contextual depth required to comprehend the "why" and "how" of ESG integration. Through this dual strategy, our results are not merely statistically robust but also based on the nuanced realities of corporate decision-making.

Limitations of Methodology

Though every effort has been made to ensure methodological rigor, there are some limitations that need to be recognized. ESG disclosures in India continue to develop, and there is heterogeneity in reporting quality and frequency among firms. Not all companies report in a rigidly following global standards, which occasionally creates inconsistencies in data comparability. Additionally, the sample size—though being diverse—is limited to six companies, which limits the generalizability of statistical results to the whole corporate universe.

Overall, the selected methodology finds a balance between empirical objectivity and strategic foresight, unifying the force of data with the depth of actual corporate histories. Through a combination of hybrid case study and statistical methods, the research aims to provide an in-depth, evidence-based understanding of how ESG is redefining the ethos and economics of contemporary Indian enterprise. Methodology

The research methodology embraced in this work combines both quantitative and qualitative measures, with rigorous emphasis on hard-case studies and statistical analysis in order to gather significant insights regarding the ESG practices of the six large-scale Indian companies: NHPC, NTPC, JSW Group, Reliance Industries, Tata Power Ltd., and Adani Power Ltd. These companies were chosen based on their level of prominence in India's energy and industrial sectors, the diversity of their ownership pattern (public sector vs private sector), and their public disclosure of ESG information.

Data Collection

This study's data was gathered through secondary research, which included a structured analysis of the BRSR, Financial, and Annual reports of the companies for the years 2021-22, 2022-23, and 2023-24. The primary research method for this study was primary research. In order to guarantee the genuineness and dependability of the information that was gathered, these reports were accessed through the official websites of the companies.

In order to identify trends and patterns concerning the financial and non-financial parameters of the companies, the study concentrated on the annual reports of the years that were mentioned above. With the purpose of conducting the analysis, key financial indicators, disclosures regarding sustainability, and data pertaining to employees were extracted. For the purpose of making comparisons and interpretations of the data easier, spreadsheets were used to organize the data in a systematic manner.

There was a lot of attention paid to making sure that all of the data sources were appropriately cited and that they were pertinent to the subject of the study.

Summary:

- 6 companies- Adani Power, Tata Power, Reliance Industries, JSW Steel, NTPC, NHPC
- 25 ESG parameters were considered.
- Sample contained data for the three financial years viz. 2021-2022, 2022-2023, and 2023-2024.

ESG Scores

The calculation of the ESG Scores for our selected companies have been done on the basis of the number of ESG parameters disclosed by each company in their Annual, Financial and BRSR reports across the financial years 2021-2022, 2022-2023 and 2023-2024. The highest attainable score is **25**, which is possible when information regarding all the required ESG parameters have been disclosed.

1. TATA POWER:

23-24	22-23	21-22
19/25 PARAMETERS	16/25 PARAMETERS	15/25 PARAMETERS

2. ADANI POWER:

23-24	22-23	21-22
19/25 PARAMETERS	18/25 PARAMETERS	21/25 PARAMETERS

3. NHPC:

23-24	22-23	21-22
4/25 PARAMETERS	5/25 PARAMETERS	5/25 PARAMETERS

4. NTPC:

23-24	22-23	21-22
23/25 PARAMETERS	23/25 PARAMETERS	22/25 PARAMETERS

5. RELIANCE INDUSTRIES:

23-24	22-23	21-22
15/25 PARAMETERS	15/25 PARAMETERS	11/25 PARAMETERS

6. JSW STEEL:

23-24	22-23	21-22
13/25 PARAMETERS	13/25 PARAMETERS	13/25 PARAMETERS

Following is the grading for the ESG score attained by a company:

SCORE	GRADE
< 10	N/A
10 - 15	Low
15 - 20	Medium
20 - 25	Excellent

The scores obtained were:

FOR 2023-2024:

COMPANY	SCORE	GRADE
NTPC	23	Excellent
Adani Power	19	Medium
Tata Power	19	Medium
Reliance Industries	15	Medium
JSW Steel	13	Low
NHPC	4	N/A

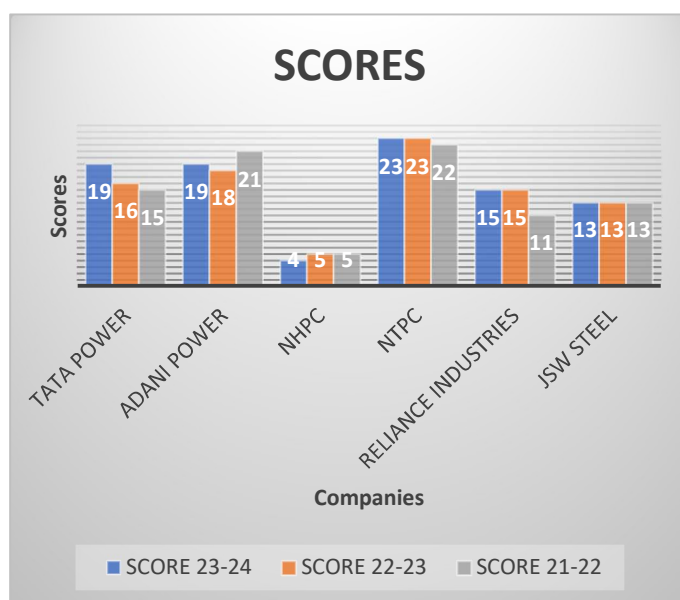
FOR 2022-2023:

COMPANY	SCORE	GRADE
NTPC	23	Excellent
Adani Power	18	Medium

Tata Power	16	Medium
Reliance Industries	15	Medium
JSW Steel	13	Low
NHPC	5	N/A

FOR 2021-2022:

COMPANY	SCORE	GRADE
NTPC	22	Excellent
Adani Power	21	Excellent
Tata Power	15	Medium
Reliance Industries	13	Low
JSW Steel	11	Low
NHPC	5	N/A



Data Analysis and Results

How data analytics benefit ESG reporting and monitoring:

Firstly, data analytics helps businesses gather ESG data from disparate sources, such as internal teams, external suppliers, and various databases that may store ESG-related data. Next, data analytics can identify and correct any data errors, inconsistencies and missing values.

Data analytics can now derive meaning from the huge volume of data, which is mostly unstructured and generated from various sources. It can also help a company spot trends and identify specific areas of improvement. For example, data analytics can help a company measure energy consumption in operations and identify functions where you are using more energy than required.

Likewise, data analytics can help draw a comparison between the energy consumption of a company to its peers in the industry. Such benchmarking can further identify areas where you can improve performance.

With real-time data analytics, companies can continuously monitor their ESG metrics and get immediate feedback on their initiatives. This ongoing assessment allows for quick adjustments and enhances the agility of ESG strategies.

Data analytics not only simplifies the process of ESG reporting but also elevates the quality and impact of the insights derived from it. As companies increasingly adopt sophisticated analytics, the depth and influence of ESG reporting are expected to expand, driving better sustainability outcomes and enhanced corporate performance.

Challenges in Integrating Data Analytics into ESG Reporting:

Integrating data analytics into ESG reporting, while beneficial, presents several challenges and considerations that vary depending on the size and resources of an organization. Here's an exploration of these challenges and specific considerations for small and medium enterprises (SMEs) versus larger corporations.

1. Cost

The financial investment required to implement data analytics can be significant. Advanced analytics software, along with the hardware needed to support such systems, can be expensive. For SMEs, this cost can be a major barrier, as they often operate with limited budgets that may not accommodate high-end analytical tools. In contrast, larger corporations might have more financial flexibility but must still justify the ROI on expensive analytics solutions, ensuring they align with long-term ESG and financial goals.

2. Complexity

The complexity of data analytics systems can be daunting. These systems often require sophisticated setup, maintenance, and management. For SMEs, the technical complexity may deter utilization due to smaller IT teams or less technical expertise. Larger corporations, while typically better equipped with IT resources, also

face challenges in scaling analytics solutions across different departments and global operations, requiring standardized procedures and training to ensure effective implementation.

3. Required Expertise

Data analytics requires a certain level of expertise to effectively interpret data and translate it into actionable insights. Both SMEs and larger companies may struggle with finding and retaining skilled personnel capable of managing and analyzing ESG data effectively. For SMEs, the talent gap can be more pronounced due to less competitive salaries or less visibility in the job market. Larger companies, although they may attract talent more easily, face the challenge of integrating expertise into a cohesive corporate strategy that aligns with ESG objectives.

By integrating advanced analytical techniques, organizations can achieve greater accuracy and reliability in their ESG data, gain deeper insights into sustainability practices, and make more informed strategic decisions. This approach not only streamlines the reporting process but also elevates the overall impact of ESG initiatives.

As the landscape of corporate responsibility continues to evolve, the need for robust data analytics in ESG reporting cannot be overstated. These tools offer the potential to transform complex datasets into actionable insights, enabling businesses to meet both regulatory requirements and stakeholder expectations more effectively.

Consistency of the data procured:

Comparing the data extracted from different companies' BRSR for different financial years, we found that the data were inconsistent.

Proceeding with the available data, we found that:

1. **Tata** could provide data on 7/11 environmental parameters, 4/5 social parameters and 6/9 governance parameters.
2. **Adani** could provide 8/11 environmental parameters, 5/5 social parameters and 7/9 governance parameters.
3. **Reliance** could provide 5/11 environmental parameters, 4/5 social parameters and 5/9 governance parameters.
4. **NTPC** could provide 11/11 environmental parameters, 4/5 social parameters and 7/9 governance parameters.
5. **NHPC** could provide 1/11 environmental parameters, 1/5 social parameters and 3/9 governance parameters.
6. **JSW** could provide on 5/11 environmental parameters, 3/5 social parameters and 5/9 governance parameters.

From the available data, we could also conclude that **NTPC** provided good data as compared to the other 5 companies, while **NHPC** didn't disclose much of their data which made the analysis and comparison difficult.

Adani provided data on capacity utilization only for FY 21-22.

Data on grievances reported was disclosed by **NTPC** for the FY 22-23 and 23-24 only and by **Reliance** for the FY 21-22 and 22-23.

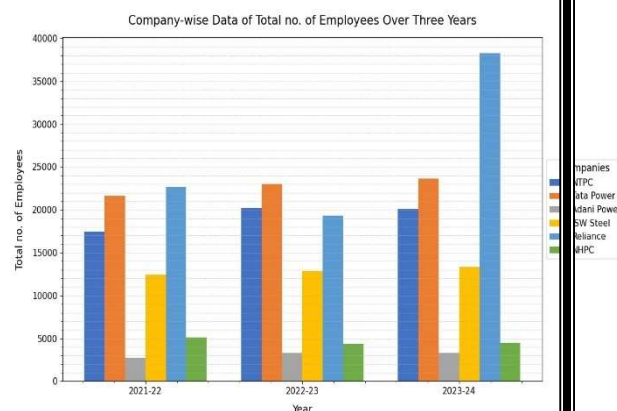
Analysis of the Data procured:

Upon analysis of the data provided by the companies in their BRSR, it was evident that –

For social parameters:

a. Total number of workers-

While, **Tata** hired more people with every financial year and had employed 8.7% more workers in FY 23-24 as compared to FY 21-22, **Adani** employed 20.7% more workers in FY 23-24 as compared to FY 21-22. **Reliance** witnessed a 14.59% decrease in employment for FY 22-23 as compared to FY 21-22, but this changed for the next FY and it accounted for an increase of 18,906 for FY 23-24 as compared to FY 22-23. **NTPC** hired 15.89% more workers in FY 22-23 as compared to FY 21-22 but in FY 23-24, there was a slight decrease in their workforce. **NHPC** witnessed a significant decrease in the number of employees with 5092 employees in FY 21-22, 4380 in FY 22-23 and 4461 in FY 23-24. **JSW Steel** increased its number of employees from 12398 in FY 21-22 to 12856 in FY 22-23 and 13301 in FY 23-24.



b. CSR (Corporate Social Responsibility) Beneficiaries Number-

Tata consistently reached out to more people and helped them. The number increased from 13.67 lakh in FY 21-22 to 37 lakh in FY 22-23 and 42.17 lakh in FY 23-24. **Adani** benefitted 37 lakh people in FY 21-22, 8.8 lakh people in FY 22-23 and 23 lakh people in FY 23-24. Even though **Reliance** did not provide data on the number of CSR beneficiaries for FY 21-22, it had reached out to 6.9 crore people in FY 22-23 but to only 64 lakh people in FY 23-24. While we do not have much information on how many people **NTPC** reached out to in FYs 21-22 and 22-23, it was reported that **NTPC** had benefitted 16 lakh people in FY 23-24. We already noted that **NHPC** has not provided consistent data, so there is no evidence of the number of CSR beneficiaries in its BRSR. However, **JSW Steel** benefitted 7.9 lakh people in FY 21-22 and 22-24 but 16 lakh people in FY 23-24.

c. CSR Expenditure:

Every company must spend at least 2% of their average net profits made during the immediately preceding three financial years as per its CSR policy.

Tata increased its budget and hence the share of CSR expenditure in its budget also increased from 32.8 cr in FY 21-22 to 50 cr in FY 22-23 and 58 cr in FY 23-24. Eventually they reached out to more people and hence spent more on the society with every financial year. **Adani** had initially reduced its CSR expenditure from 26.56 cr in FY 21-22 to 16.70 cr in FY 22-23 but increased it in FY 23-24 to 57.67cr. **Reliance** did not disclose how much it had spent on CSR in the FY 21-22. However, it had spent more on CSR in FY 23-24 (35cr) as compared to 2.3 cr in FY 22-23. **NTPC** dedicated 356.72 cr to CSR projects in FY 21-22, 315.32 cr in FY 22-23 and 425.76 cr in FY 23-24. **NHPC** BRSR, having inconsistent data, did not disclose how much they had spent on CSR projects in any of the financial years. **JSW Steel** did not provide the data on CSR expenditure as well.

d. LTIFR (Loss Time Injury Frequency Rate) of employees:

Tata recorded zero LTIFR of employees for FYs 21-22 and 22-23 but 0.06 for FY 23-24. **Adani** recorded 0.03 LTIFR of employees for FY 21-22, 0.14 for FY22-23 but zero for FY 23-24. **Reliance** did not disclose information on this parameter for FY 21-22. However, it recorded 0.21 LTFIR in FY 22-23 and witnessed an increase of 0.08 for the same in FY23-24. **NTPC** recorded a consistent decrease in its LTIFR of employees, from 0.13 in FY 21-22 to 0.12 in FY 22-23 and 0.1 in FY 23-24. **JSW Steel** witnessed a slight increase in LTIFR of its employees from 0.32 in FY 21-22 to 0.33 in FY 22-23. But this decreased to 0.11 in FY 23-24. **NHPC** did not provide information related to this in its BRSR.

For Environmental Parameters:

Out of 11 parameters, 4 most important parameters were picked up for the analysis to make it more precise and concise.

a) Energy Consumption-

Tata has consumed 27,80,72,239 GJ of energy in FY 21-22, 40,89,48,131 GJ in FY 22-23 and 50,54,40,434 GJ in FY 23-24. They have only increased their energy consumption over the years. However, their energy consumption increased by 47.06% in FY 22-23 as compared to FY 21-22 and then for the next financial year, their energy consumption increased by 23.5 % in FY 23-24 as compared to FY 22-23. This means that they increased their energy consumption at a lower rate.

Adani consumed 52,69,90,357 GJ of energy in FY 21-22, 55,05,24,092 GJ in FY 22-23 and 81,84,44,562 GJ of energy in FY 23-24. Even though they showed a lower rate of increase in energy consumption in FY 22-23 as compared to FY 21-22, they witnessed a large increase in this parameter in FY 23-24. The reason for such a large increase is unknown.

Reliance witnessed a gradual decrease in its energy consumption from 48,58,01,414 GJ in FY 21-22 to 47,35,90,585 GJ in FY 22-23 and 46,42,00,812 GJ in FY 23-24.

NTPC accounted for 14% increase in its energy consumption in FY 22-23 as compared to FY 21-22 and a 16.41% in energy consumption in FY 23-24 as compared to FY 22-23.

NHPC stated in their BRSR that their energy consumption was 24494 GJ in FY 21-22, 24907 GJ in FY 22-23 and 21779 GJ in FY 23-24. They consumed less energy as compared to the other companies and reduced their consumption every year but with a slight increase in FY 22-23 as compared to FY 21-22.

JSW Steel consumed 10.65 % more energy in FY 22-23 as compared to FY 23-24 and 4.71% more energy in FY 23-24 as compared to FY 22-23.

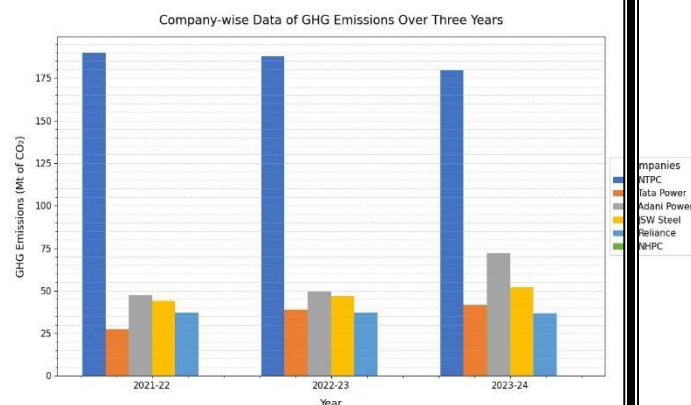
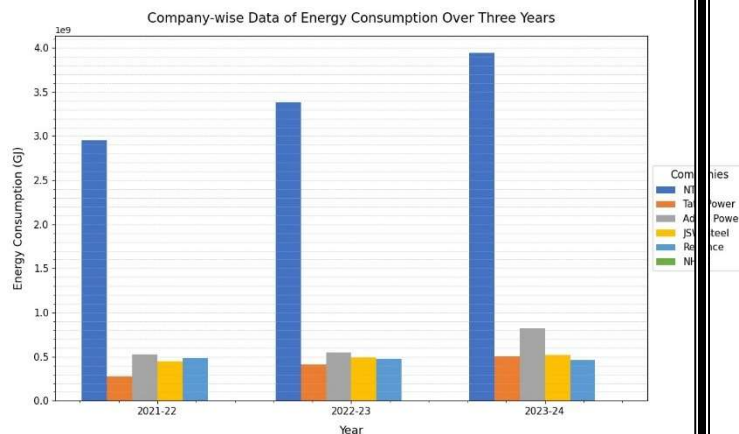
So, most companies tried in their own ways to lower their energy consumption. However, Reliance was more impressive for that matter.

b) GHG (Green House Gas) Emission –

Tata produced 40.74% more GHG in FY 22-23 as compared to FY 21-22 and 6.9% more GHG in FY 23-24 as compared to FY 22-23. This proves that they have tried to reduce the emission of GHG over FY 22-23 and FY 23-24.

Adani witnessed a 3.8% increase in GHG emission in FY 22-23 as compared to FY 23-24 and a 45.48% increase in FY 23-24 as compared to FY 22-23. This shows that they have accounted for a significant increase in GHG emission with no step towards judicious use of the resources.

Reliance emitted 37.2 million tCO₂ in FY 21-22, 37 million tCO₂ in FY 22-23 and 36.9 million tCO₂ in FY 23-24 which shows that they tried to reduce their GHG emissions over the years.



NTPC's GHG emissions reduced by a little margin over the three FY.

NHPC could not provide any information on GHG emissions.

JSW Steel marked a 6.17% increase in GHG emissions in FY 22-23 as compared to FY 21-22 and a 11% increase in FY 23-24 as compared to FY 22-23.

c) Coal Consumption –

Tata consumed 5.9% more coal in FY 22-23 as compared to FY 21-22 and 22.7% more coal in FY 23-24 as compared to FY 22-23. We can infer that they relied heavily on conventional sources of energy.

Adani didn't disclose how much they consumed coal in FY 22-23. However, from the available data we could conclude that their coal consumption increased by 45.25% in FY 23-24 as compared to FY 21-22 which is quite high.

NTPC accounted for 55.01% increase in coal consumption in FY 22-23 as compared to FY 21-22 and a 0.6% increase in FY 23-24 as compared to FY 22-23. This shows that even though their coal consumption had a significant increase in FY 22-23, they tried to work with the same amount in FY 23-24, as they did in FY 22-23 but with a slight increase.

Companies like **NHPC** and **JSW Steel** did not disclose information on this parameter. Hence, the analysis could not be done.

d) Fly Ash –

Tata used 100% of fly ash in FY 21-22, 94.6% in FY 22-23 and 97.40% in FY 23-24.

Adani used 86.70% of fly ash in FY 21-22, 100% in FY 22-23 and 89% in FY 23-24.

NTPC used 100 % in all three FYs.

Reliance, NHPC and **JSW Steel** did not disclose their data on fly ash utilization.

NTPC proves to be best in this parameter.

For Governance Parameters:

Out of 9 parameters, 4 were considered for the analysis.

a) Revenue-

Tata generated 31.60% more revenue in FY 22-23 as compared to FY 21-22 and 9.8% more revenue in FY 23-24 as compared to FY 22-23.

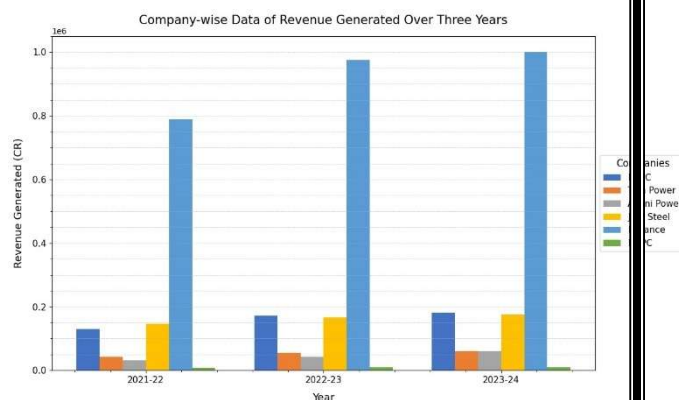
Adani generated 35.83% more revenue in FY 22-23 as compared to FY 21-22 and 40.05% more revenue in FY 23-24 as compared to FY 22-23. It generated more revenue over the years as compared to **Tata**.

Reliance accounted for 23.59% more generation of revenue in FY 22-23 as compared to FY 21-22 and 2.59% more in FY 23-24 as compared to FY 22-23.

NTPC witnessed 32.76% more revenue generation in FY 22-23 as compared to FY 21-22 and 4.6% more in FY 23-24 as compared to FY 22-23.

NHPC reported an 8.7% increase in its revenue in FY 22-23 as compared to FY 21-22 and a 0.2% decrease in its revenue in FY 23-24 as compared to FY 22-23.

JSW Steel marked a 13.38% increase in its revenue in FY 22-23 as compared to FY 21-22 and a 5.4% increase in its revenue in FY 23-24 as compared to FY 22-23.

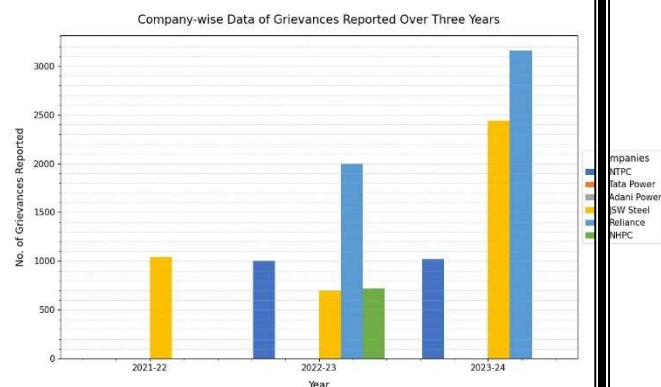


Out of all companies, Adani marked a consistent increase in its revenue.

b) Human Rights Violation Report –

Tata reported 3 cases for sexual harassment and 5 for discrimination in FY 22-23 and 3 cases for sexual harassment along with 9 cases for discrimination in FY 23-24. They didn't disclose any information related to this in their BRSR for FY 21-22.

Adani did not report any cases of sexual harassment or discrimination in any one of the financial years except for 1 case of sexual harassment in FY 23-24.



Reliance, NTPC, NHPC and JSW Steel did not report any cases of sexual harassment or discrimination in any one of the financial years.

c) Spend on R&D (Research and Development) –

Tata had spent 76.72% more on R&D in FY 22-23 as compared to FY 21-22 and 12.33% more in FY 23-24 as compared to FY 22-23.

Adani had spent 33.8% less on R&D in FY22-23 as compared to FY 21-22 and just 1.08 cr on R&D in FY 23-24 as compared to 99.3 cr in FY 22-23.

Reliance's expenditure in R&D was 15.06% more in FY 22-23 as compared to FY 21-22 and 15.39% more in FY 23-24 as compared to FY 22-23.

NTPC had spent 220 cr in FY 21-22 and just 40 cr more in FY 22-23. However, there wasn't any information on their expenditure on R&D for FY 23-24.

NHPC and JSW Steel did not disclose any data on their expenditure on R&D for any one of the FYs in BRSRs.

d) Profit –

Tata marked a 31.60% rise in its profit in FY 22-23 as compared to FY 21-22 and a 12.33% rise in FY 23-24 as compared to FY 22-23.

Adani's profit rose by 118% in FY 22-23 as compared to FY 21-22 and it rose by 94% in FY 23-24 as compared to FY 22-23.

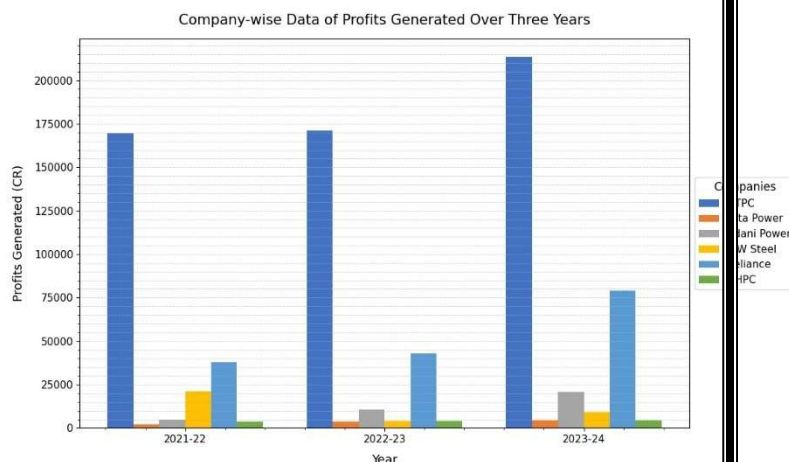
Reliance witnessed a 13.39% increase in its profit in FY 22-23 as compared to FY 21-22 and 83.69% rise in its profit in FY 23-24 as compared to FY 22-23.

NTPC marked a 0.94% increase in FY 22-23 as compared to FY 21-22 and a 24.59% increase in its FY 23-24 as compared to FY 22-23.

NHPC marked a 20.16% increase in its profit in FY 23-24 as compared to FY 21-22.

JSW Steel marked a 57.14% decrease in its profit in FY 23-24 as compared to FY 21-22.

Comparing all the parameters we can conclude that even though **Adani** and **NTPC** had the most consistent data in financial parameters and had a good performance in such parameters but **Adani** consumed energy way more than what the other companies did. They didn't spend much on R&D either. **NHPC** had the most inconsistent data. So, the analysis for it became difficult. Profit of **JSW Steel** marked an overall decrease of 57.14%.



Recommendations-

For Reliance:

Data for key environmental parameters involving coal consumption were not revealed, due to which the actual impact on the environment could not be primarily assessed.

For JSW:

The data for key environmental parameters weren't revealed by JSW either hence the analysis became troublesome. The ratio of the number of grievances reported to the total number of employees was significantly higher than that of many companies, so employee welfare should be taken care of.

For Tata:

Data for coal consumption implied that they relied heavily on conventional sources of energy instead of non-conventional sources of energy. Hence, they should work on it and try to shift to non-conventional sources of energy. They should also work on human rights and bring out policies to lower the numbers of cases of sexual harassment and discrimination.

For Adani:

Adani should spend more on CSR and reach out to more people to benefit them and should also spend more on R&D. They should also resort to judicious use of energy by lowering their GHG emission as it significantly affects the environment. They do not use 100% of fly ash which needs to be looked upon. Only generating more revenue would not guarantee a sustainable and judicious use of resources.

For NHPC:

NHPC's ESG disclosures were scant and incoherent, making effective analysis constrained. Environmental impact, social well-being, and governance-related crucial data were not available. To increase transparency and conform to ESG norms, NHPC must enhance reporting processes, particularly with regard to energy consumption, CSR initiatives, and worker safety. Improving data systems and year-on-year consistency is vital to establish credibility and facilitate meaningful sustainability evaluation.

For NTPC:

NTPC demonstrated robust ESG disclosure, especially in energy and utilization of fly ash. The sudden spike in coal consumption during FY 22-23 is environmentally worrisome. Though CSR expenditure was high, beneficiary information was inconsistent. NTPC must emphasize adoption of renewable energy, ensure transparency in all parameters, and enhance R&D expenditure to facilitate sustainable innovation and long-term environmental stewardship.

Outcome

The implication of this study is a multi-faceted and comparative perspective of ESG (Environmental, Social, and Governance) practices since the results of this study produce various significant outcomes that reflect on both the present situation and potential future of ESG integration in Indian corporate systems.

One of the standout achievements is NTPC being identified as the ESG disclosure leader, consistently performing well in environmental, social, and governance metrics. It revealed the greatest amount of information, utilized 100% of fly ash, and had robust CSR and R&D expenditure records. Tata Power and Adani Power also showcased good efforts, especially in CSR engagement and employee increases, although their overdependence on coal and irregular fly ash usage were observed areas of concern. Conversely, NHPC turned out to be a laggard with poor ESG disclosures and lack of data consistency, thus making comparative analysis difficult.

Data analytics for ESG reporting was also emphasized by the research. Firms utilizing real-time monitoring of data and benchmarking exemplified higher reporting quality and operating responsiveness. Yet, analytics integration in ESG is restrained by cost and technical limitations, particularly for smaller companies. This indicates an intention-action gap that necessitates enhanced digital infrastructure and capacity development.

Another result relates to the financial advantages of ESG implementation. Firms with solid ESG systems, such as Tata Power and NTPC, were seen to have more reputational capital, investor trust, and even profitability. For example, companies that had raised spending in R&D demonstrated positive indications of growth through innovation. At the same time, others like JSW Steel, which saw a significant fall in profits and little ESG improvement, face financial underperformance from unsustainable practices.

Additionally, the report reveals sectoral insights. Firms with high energy consumption, especially in fossil-fuel-based electricity generation, still struggle with GHG emissions reduction and cleaner energy integration. Reliance Industries and Adani Power, even with robust financials, have to strategically focus on decarbonization in alignment with the world's ESG expectations.

In totality, the result of this research leans towards a dualistic ESG terrain of Indian corporates—characterized by encouraging advancements, structural bottlenecks, and sectoral variation. The report underscores the need for instant standardization of ESG reporting, greater regulatory control, and deliberate investment in sustainable practices. With ESG increasingly defining investor choices and company valuations, Indian businesses have to transition from compliance-oriented thinking toward value-oriented ESG approach based on long-term sustainability, innovation, and transparency

Policy Recommendation

On the basis of this all-encompassing assessment of ESG practices of top Indian corporates, a number of pressing policy suggestions emerge to bolster India's ESG environment. To begin with, the ambit of obligatory ESG disclosures ought to be broadened from the top 1000 listed companies to encompass all large and mid-cap entities, with a well-defined compliance plan for smaller businesses utilizing simplified formats such as BRSR Lite. For ensuring comparability and uniformity, the government, jointly with SEBI and the Ministry of Corporate Affairs, must implement standardized ESG metrics and data disclosure templates based on international standards like GRI, SASB, and TCFD. An e-platform portal for centralized, publicly available ESG reporting with machine-readable data sets would ensure data transparency, reduce inconsistencies—as in instances like NHPC's lack of environmental data—and enhance investor confidence. To incentivize actual commitment rather than just box-ticking, a graded system of incentives and penalties should be institutionalized, with tax incentives or simpler access to green financing for top-performing companies, and tighter scrutiny or reputational risk for non-compliers. Technical capacity remains a major challenge, especially among public and mid-sized firms; therefore, targeted training programs, national ESG certification schemes, and a helpdesk portal must be launched to assist companies in integrating ESG principles effectively. Public-private partnerships with academic institutions and think tanks can foster skill-building and awareness at the grassroots. In addition, ESG must be applied to the specifics of each industry; therefore, industry associations like CII, FICCI, and ASSOCHAM must be incentivized to create sectoral ESG benchmarks that outline expectations for environmental regulation, social responsibility, and governance best practice—for example, power sector benchmarks could serve to bring companies like Adani and Tata in line with leaders like NTPC in fly ash use and GHG reduction. Incorporating ESG performance into public procurement, sovereign credit analyses, and credit rating processes can further institutionalize sustainability in the Indian economy, with the penalty for non-compliance being loss of financial and reputational capital. Lastly, to enable long-term transition, policymakers need to move the national ESG narrative from compliance to competitiveness—where businesses are enabled to view ESG as not a regulatory burden, but a strategic driver of innovation, resilience, and investor confidence. Through this multi-faceted policy strategy—grounded in standardization, accountability, training, incentives, and sectoral customization—India will be able to develop a strong ESG ecosystem that fosters both sustainability and inclusive growth aligned with national and global development objectives.

Conclusion

This study serves as a practical tool for assessing the initial sustainability status of organizations by integrating widely recognized sustainability indicators from global indices, standards, and frameworks. It simplifies complex data into a composite index, allowing for easy comparison and enabling cross-sectional analysis, as demonstrated in this research. The environmental, social, and governance disclosure metrics are consolidated into a single value that provides meaningful insights into corporate sustainability performance.

The study contributes significantly to the existing body of literature on business sustainability performance by selecting appropriate parameters relevant to the Indian context. Its uniqueness lies in adapting global sustainability concepts to evaluate Indian enterprises, facilitating comparative studies across similar industry sectors.

The research also explores whether significant differences exist in sustainability performance among companies within the same industry category in India. Data collected over three consecutive years indicates noticeable differences in performance trends. Large-cap, internationally active firms have demonstrated superior sustainability practices, often supported by robust reporting systems and comprehensive sustainability disclosures aimed at minimizing the negative impact of their operations.

Findings suggest that larger corporate entities outperform their small- and medium-sized counterparts in terms of sustainability reporting, management, and accountability. The Ministry of Corporate Affairs, in collaboration with SEBI, has introduced the Business Responsibility and Sustainability Report (BRSR) framework to enhance sustainability practices in Indian companies. This study highlights the potential of index-based methodologies for sustainability analysis and suggests the incorporation of advanced data analytics tools in future research.

However, the scope of this study was constrained by time and the limited availability of primary data. Future research could benefit from detailed unit-level analyses across various business types. Given the complexity of real-world business environments, the current set of indicators could be further refined based on organizational size, sector, and operations. Additionally, future studies should focus on developing new, context-specific criteria and guidelines instead of solely relying on existing conceptual frameworks for evaluating sustainability performance.

References

This section contains the raw data for the selected companies that was compiled for the research, followed by the list of references used for this project:

1. Reliance Industries:

YEAR	SPECIFIC WATER CONSUMPTION (L/KWh)	SPECIFIC COAL CONSUMPTION (KG/KWh)	COAL CONSUMPTION (MT)	ENERGY CONSUMPTION (GJ)	ENERGY INTENSITY (KJ/KWh)	GHG EMISSION (MILLION TONNES CO2)	GHG INTENSITY (Tco2/KWh)	ENERGY SAVING (million GJ)	ENERGY GENERATION (BU)	RENEWABLE GENERATION (BU)	FLY-ASH UTILIZATION (%)
2023-2024				464200812	0.0000807	36.9	0.00000655	5280000			
2022-2023				473590585	0.00008377	37	0.00000671	2530000			
2021-2022				485801414	0.00010446	37.2	0.00000819	2110000			

YEAR	TRAINING COVERAGE (HOURS/EMPLOYEE/YEAR)	TOTAL EMPLOYEES	CSR BENEFICIARIES	CSR SPEND (INR)	LTIFR OF EMPLOYEES
2023-2024		38244	6434688	352821082	0.29
2022-2023		19338	69500000	23241353	0.21
2021-2022		22642			

YEAR	REVENUE (crore)	GRIEVANCES REPORTED	HUMAN RIGHTS VIOLATION REPORT	ANTI-CORRUPTION AND ANTI-BRIBERY REPORTED	INVESTMENT FOR DEVELOPMENT (CRORE)	SPEND ON R&D (CRORE)	PROFIT (CRORE)	PLF (%)	CAPACITY UTILIZATION (%)
2023-2024	1000122	3157	NIL	NIL		3463	79020		

2022-2023	974864	1998	NIL	NIL		3001	43017		
2021-2022	788743		NIL	NIL		2608	37937		

2. JSW Steel:

YEAR	SPECIFIC WATER CONSUMPTION(L/KWh)	SPECIFIC COAL CONSUMPTION(KG/KWh)	COAL CONSUMPTION(MT)	ENERGY CONSUMPTION(GJ)	ENERGY INTENSITY(KJ/KWh)	GHG EMISSION (MILLION TONNES CO2)	GHG INTENSITY (Tco2/KWh)	ENERGY SAVING (GJ)	ENERGY GENERATION(BU)	RENEWABLE GENERATION(BU)	FLY-ASH UTILIZATION(%)
2023-2024				517690735	0.000387	52.106566	0.000891	16640000			
2022-2023				494384430	0.00038	46.941683	0.000846	92030000			
2021-2022				446782474	0.00038	44.21131	0.038	15080000			

YEAR	TRAINING COVERAGE(HOUR/EMPLOYEE/YEAR)	TOTAL EMPLOYEES	CSR BENEFICIARIES	CSR SPEND	LTIFR OF EMPLOYEES
2023-2024		13301	1699137		0.11
2022-2023		12856	790879		0.33
2021-2022		12398	790879		0.32

YEAR	REVENUE	GRIEVANCES REPORTED	HUMAN RIGHTS VIOLATION REPORT	ANTI-CORRUPTION AND ANTI-BRIBERY REPORTED	INVESTMENT FOR DEVELOPMENT(CRORE)	SPEND ON R&D(CRORE)	PROFIT(CRORE)	PLF(%)	CAPACITY UTILIZATION(%)
2023-	175006	2439	NIL	NIL			8973		

2024									
2022-2023	165960	699	NIL	NIL			4139		
2021-2022	146371	1042	NIL	NIL			20938		

3. TATA Power:

Parameters	FY 21-22	FY 22-23	FY 23-24
Training Coverage (hour/employee/ year)			
Total Employee No.	21636	23025	23652
CSR Beneficiaries No.	1367000	3700000	4217000
CSR Expenditure	32.8 cr	50 cr	58 Cr
LTIFR (Loss Time Injury Frequency Rate) of employees	Nil for employees + 0.15 worker	Nil for employees + 0.11 workers	0.06 Employees + 0.31 Workers
Specific Water Consumption (kl/kwh)			0.005036
Specific Coal Consumption (kg/kwh)			
Coal Consumption (MT)	20.3 MT	21.5 MT	26.4 MT
Energy Consumption (GJ)	278072239	408948131	505440434
Energy Intensity (kj/kwh)			795 GJ
GHG Emission (million tonnes CO2)	27.616	38.868	41.55
GHG Intensity (CO2/kwh)			
Energy savings (million watt hour)	1976	3451	2280
Energy generation (BU)	10,115 MW (conventional)	10,183 MW (conventional)	10,183 MW
Renewable generation (BU)	3,400 MW	3,927 MW (renewable)	4,524 MW
Fly Ash Utilisation (%)	100%	94.60%	97.40%
Revenue	42,576 cr	56,033 cr	61,542 cr
Grievances reported			
Human Rights Violation Report		3 for Sexual Harrassment and 5 for discrimination	3 for Sexual Harrassment and 9 for discrimination
Anti-corruption and anti-			Nil

bribery reported			
Investment for development (cr)			
Profit (cr)	2156 Cr	3810 Cr	4280 Cr
Spent on R&D (Cr)	13.7 cr	17.06 crore	12.21 cror
PLF (%)	87.00%	95%	95%
Capacity Utilisation (%)			

4. Adani Power:

Parameters	FY 21-22	FY 22-23	FY 23-24
Training Coverage (hour/ employee/ year)	2746 employees	3313 employees	3315 employees
Total Employee No.	2746	3313	3315
CSR Beneficiaries No.	1000000	886582	2300000
CSR Expenditure	26.56 cr	16.70 cr	57.67 Cr
LTIFR (Loss Time Injury Frequency Rate) of employees	0.03	0.14 employees + 0.02 workers	0.00 employees + 0.15 workers
Specific Water Consumption (l/kwh)	2.32 M3/MWh	2.31 m3/MWh	2.25 m3/MWh
Specific Coal Consumption (kg/kwh)			
Coal Consumption (MT)	35296922 MT		5,12,71,728.69 MT
Energy Consumption (gJ)	526990356.71	550524091.54	818444561.86
Energy Intensity (kj/kwh)	9.38 GJ/MWh	9.59 GJ/MWh	9.57 GJ/MWh
GHG Emission (million tonnes CO2)	47.672	49.492	72.005
GHG Intensity (CO2/kwh)	0.85 tCO2e/MWh	0.85 tCO2e/MWh	0.85 tCO2e/MWh
Energy savings (million watt hour)			
Energy generation (BU)			
Renewable generation (BU)	61,652.16 MWh		
Fly Ash Utilisation (%)	86.70%	100%	89%
Revenue	31686	43040.52	60281
Grievances reported	5	0	0
Human Rights Violation Report	nil	nil	1 for sexual harassment
Anti-corruption and anti-bribery reported	nil	nil	nil

Investment for development (cr)			
Profit (cr)	4911.58 Cr	10727 Cr	20829 Ccr
Spent on R&D (Cr)	150 cr	99.3 cr	1.08 Cr.
PLF (%)	52%	47.90%	64%
Capacity Utilisation (%)	74.50%		

5. NTPC:

YEAR	SPECIFIC WATER CONSUMPTION(L/KWh)	SPECIFIC COAL CONSUMPTION(KG/KWh)	COAL CONSUMPTION(MT)	ENERGY CONSUMPTION(GJ)	ENERGY INTENSITY(KJ/KWh)	GHG EMISSION (MILLION TONNES CO2)	GHG INTENSITY (gCO2/KWh)	ENERGY SAVING (MWh)	ENERGY GENERATION(BU)	RENEWABLE GENERATION(BU)	FLY-ASH UTILIZATION(%)
2023-2024	2.48	0.71	34.38	3941682421.38	3.94	179.7	834.81	534722.50	422	10.2	100
2022-2023	2.69	0.73	34.15	3385762614.07	3.39	187.8	840.75	481788.20	399.3	10	100
2021-2022	2.76	0.74	22.03	2953966791.77	2.95	189.9	843.64	470905.30	~300	10	100

YEAR	TRAINING COVERAGE(HOUR/EMPLOYEE/YEAR)	TOTAL EMPLOYEES	CSR BENEFICIARIES	CSR SPEND (INR- CRORE)	LTI OF EMPLOYEES	PUBLISHING SUSTAINABILITY REPORT
2023-2024	56	20074	1600000	425.76	0.1	Integrated Annual Report
2022-2023	55	20252		315.32	0.12	
2021-2022	54	17474		356.72	0.13	

YEAR	REVENUE	GREIVANCES REPORTED	HUMAN RIGHTS VIOLATIONS REPORTED	INVESTMENT FOR DEVELOPMENT (CRORE)	SPENT ON R&D (CRORE)	PROFIT (CRORE)	PLF (%)	CAPACITY UTILIZATION (%)	ANTI-CORRUPTION & ANTI-BRIBERY REPORTED
2023-2024	180605.13	1018		79738		213320	77	77	0
2022-2023	172651.9	1003		20922	280	171210	75.9	75.9	0
2021-2022	130047.2			250000	220	169600	70.7	70.7	1

6. NHPC:

YEAR	SPECIFIC WATER CONSUMPTION(L/ KWh)	SPECIFIC COAL CONSUMPTION(KG /KWh)	COAL CONSUMPTION(MT)	ENERGY CONSUMPTION(GJ)	ENERGY INTENSITY(KJ /KWh)	GHG EMISSION (MILLION TONNES CO2)	GHG INTENSITY (Tco2/ KWh)	ENERGY SAVINGS(MWh)	ENERGY GENERATION (BU)	RENEWABLE GENERATION (BU)	FLY-ASH UTILIZATION (%)
2023-2024	NA	NA	NA	21779		NA		NA			NA
2022-2023	NA	NA	NA	24907		NA		NA			NA
2021-2022	NA	NA	NA	24494		NA		NA			NA

YEAR	TRAINING COVERAGE(HOUR/EMPLOYEE/YEAR)	TOTAL EMPLOYEES	CSR BENEFICIARIES	CSR SPEND (INR)	LTIFR OF EMPLOYEES	PUBLISHING SUSTAINABILITY REPORT
2023-2024		4461	NA			
2022-2023		4380	NA			
2021-2022		5092	NA			

YEAR	REVENUE	GREIVANCES REPORTED	HUMAN RIGHTS VIOLATION REPORTED	INVESTMENT FOR DEVELOPMENT (CRORE)	SPENT ON R&D (CRORE)	PROFIT (CRORE)	PLF (%)	CAPACITY UTILIZATION (%)	ANTI-CORRUPTION & ANTI-BRIBERY REPORTED
2023-2024	9316					4234.74	88.19		
2022-2023	9335	719		6961		3890	88.75		
2021-2022	8581			5166		3524	77.6		

Additionally, the following sources were referred to in the making of this project:

- SEBI (Securities and Exchange Board of India): <https://www.sebi.gov.in/>
- IAS Plus (Deloitte): <https://www.iasplus.com/en/resources/sustainability/sustainability>
- Investopedia: <https://www.investopedia.com/global-reporting-initiative-7483127>
- NIFTY100 ESG: ESG Analysis on 100 Listed Companies-India INC's Readiness for Regulatory Push-2022
- BRSRs of TATA:
 - <https://www.tatapower.com/content/dam/tatapoweraemsitesprogram/tatapower/pdf-root/company-financials/annual-reports/103AnnualReport-2021-22.pdf> ,
 - <https://www.tatapower.com/content/dam/tatapoweraemsitesprogram/tatapower/pdf-root/company-financials/annual-reports/104AnnualReport-2022-23.pdf> ,
 - <https://www.tatapower.com/content/dam/tatapoweraemsitesprogram/tatapower/pdf-root/company-financials/annual-reports/tatapower-annual-report-2023-2024.pdf>
- BRSRs of Adani: [Investor-Downloads](#)
- BRSRs of NTPC: <https://ntpc.co.in/sustainability/reports-and-publications>
- BRSRs of NHPC: https://www.nhpcindia.com/assests/pzi_public/gallery/17183553740.pdf
https://www.nhpcindia.com/welcome/sub_page/240.html
<https://www.nhpcindia.com/welcome/page/412.html>
- Annual Reports of NTPC: <https://ntpc.co.in/investors/annual-reports>
- Annual Reports of NHPC: https://www.nhpcindia.com/assests/pzi_public/gallery/1672224798.pdf
- NSE Archives:
 - https://nsearchives.nseindia.com/corporate/NTPC1_06082024115916_NTPCSE_BRSR_2024.pdf
 - https://nsearchives.nseindia.com/corporate/NHPC_10082023181309_BRSR_202223.pdf